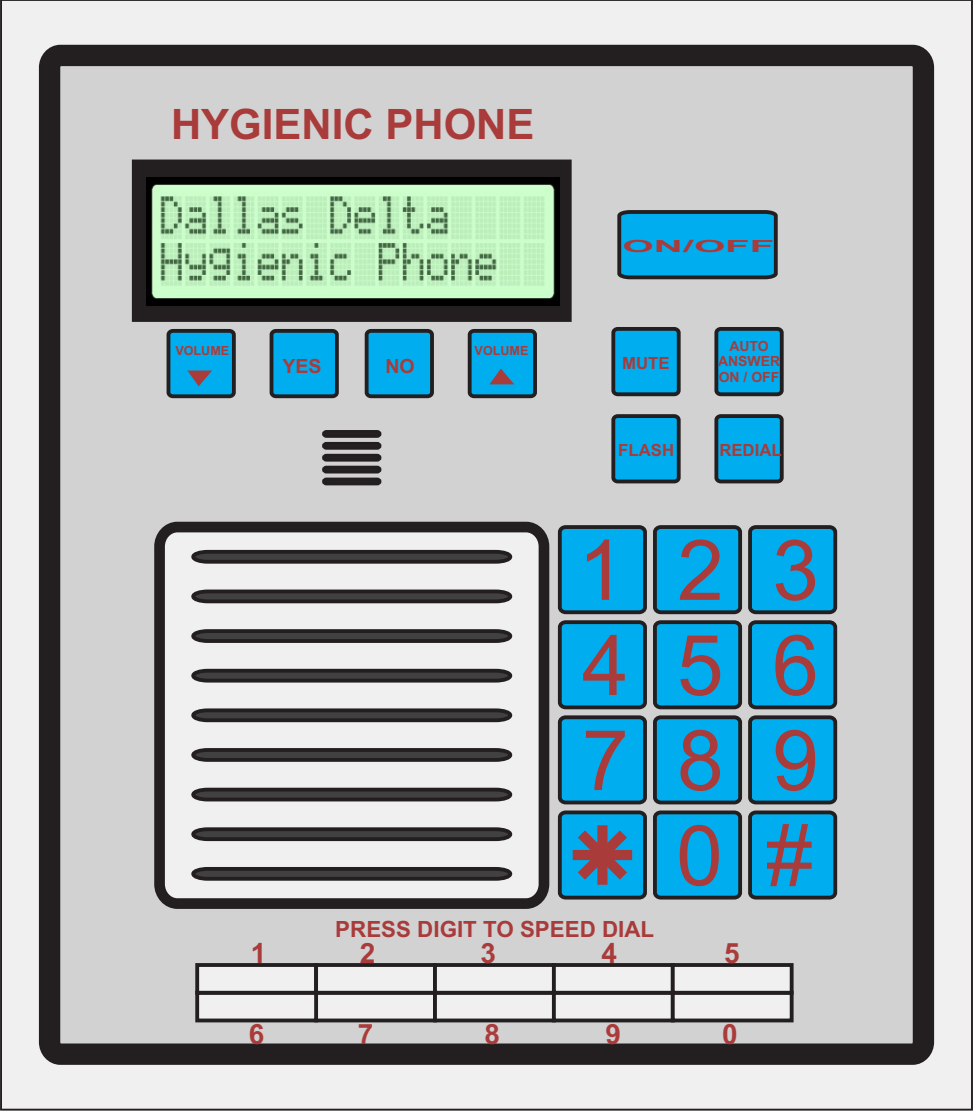




VoIP Hygienic

S-11 User's Manual

VoIP Hygienic Phone S11



DallasDelta

Corporation Pty. Ltd.

Custom Communication Solutions

Dallas Delta Corporation Pty. Ltd. is an established and respected provider of specialist communications equipment.

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1.0 Introduction

Thank you for choosing Dallas Delta products. We hope that our product will meet the demands of your application needs and it will give you many years of trouble free use.

This product was designed primarily for hygienic area applications to make it easy for people working in clean rooms and operating theatres to take and make telephone calls with the least inconvenience.

The keys were made larger than normal to take into account that the user may be gloved and the material used for the panel was chosen to be impervious and hard-wearing and easy to be cleaned.

Many features have been incorporated into this model making the hygienic phone user friendly, reliable, and safe.

Features such as Auto Answer and Auto Hangup allows for less interaction from the user, and the Soft Ramping Ringer avoids the sound shock from a sudden loud ringing during a critical procedure.

This document outlines the units Features, Operation, Programming instructions and Installation procedures.

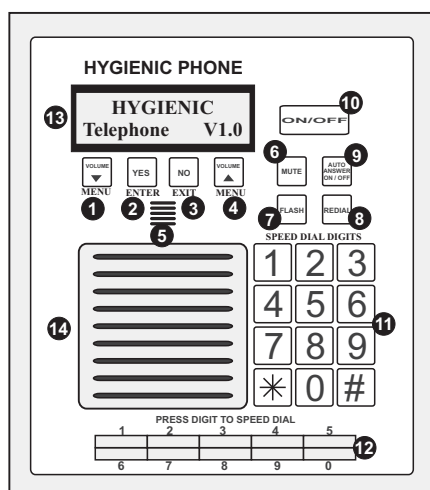
2.0 How to Use this Manual

This manual is set out with the intended use of both installers and end users in mind. It will give you the ease and quick turn-around time of installation required to get the job done efficiently and will also give you a very detailed understanding of the unit you are installing and putting into real-world use.

It is very important that installation of this unit is carried out by a qualified personnel to avoid damage of the unit and correct functionality of the product. It will also ensure that safety requirements are met for everyday use of this electrical product.

When installing this product for the first time, it is critical to follow the quick start guide to begin with. This will ensure that your new product is up and running in a very simple and easy manner with as minimal time as possible.

It is then recommended that the manual be read from cover to cover to allow you to fully understand the capabilities of your product and be able to obtain the most of this unit.



Panel Layout

- 1 Volume down/Menu Down
- 2 Yes / Enter
- 3 No / Cancel
- 4 Volume Up / Menu Up
- 5 Microphone position
- 6 Mute / Mute on Answer / PABX Pause entry
- 7 Flash / Transfer
- 8 Redial
- 9 Auto Answer function ON / OFF
- 10 ON /OFF hook
- 11 Number entry keypad
- 12 Speed dial memory directory
- 13 LCD display

To make a telephone call:

Press the **ON/OFF** (10) button then dial the number via the keypad (11).
 To dial from a Speed Dial location just press the assigned keypad digit (11).
 To Redial press the **REDIAL** (8) button.
 To hang up press the **ON/OFF** (10) button again.

To change the Volume:

To increase the speaker volume during a call, if not already set to maximum, press the volume **UP** key (4), to decrease press the volume **DOWN** key (1).

To mute a call in progress:

To mute the microphone during a call press **MUTE** (6), and again to release.

To transfer a call:

Press the **FLASH** button(7), and then follow the PBX standard procedures.

To Auto Answer:

The **AUTO ANSWER**(9) key enables or disables the Auto Answer function.

The **AUTO HANGUP** feature is always enabled and it will hang up the call after a number of busy tones.

The **Mute Microphone on Auto Answer** facility can be enabled or disabled by pressing the **MUTE** key (6) when off line.

Relay Activation:

If on board relays are fitted, they may be activated during conversation by pressing the **relay code** for each relay fitted. Each relay has a different code. When the relay is activated it will stay on for the period set in RelayOnTimer (system section).

The relay may also be set to activate when ever to phone is in use. (see the configuration section 'Relay In Use' option).

4.0 Quick Start Guide

4.1 Getting Started - Basic Procedures

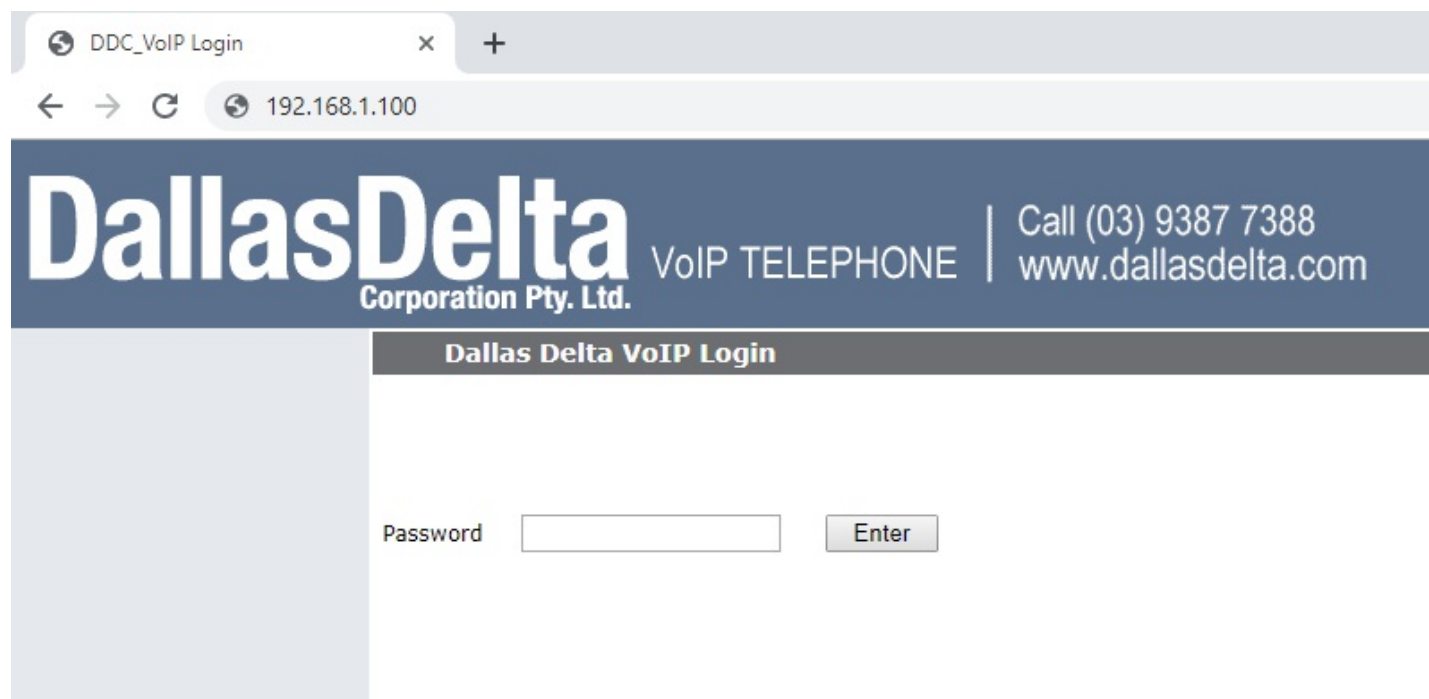
There are a few basic procedures that will need to be completed in order to get your hygienic phone working. Please follow these simple procedures in order before attempting to use your phone:

- > Log on to your phone using a modern web browser
- > Enter basic 'Network' parameters
- > Enter basic 'Systems' parameters
- > Enter basic 'SIP Proxy' parameters
- > Program your call numbers for speed dialling

Below are very simple to follow step-by-step procedures to getting your hygienic phone up and running:

4.2 Logging On to the Home Page

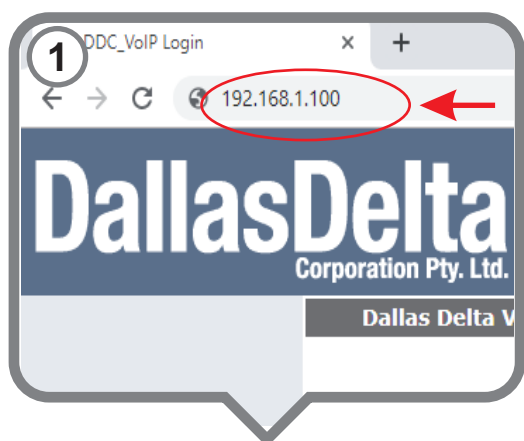
In order to begin setting up your hygienic phone you must first log into the unit via a web browser (such as Google Chrome, Firefox, etc.). Log into your hygienic phone by typing in the IP address of the phone in the address bar.



The screenshot shows a web browser window with the title 'DDC_VoIP Login'. The address bar contains the IP address '192.168.1.100'. The main content area has a dark blue header with the 'DallasDelta' logo, 'VoIP TELEPHONE', and contact information: 'Call (03) 9387 7388' and 'www.dallasdelta.com'. Below the header is a section titled 'Dallas Delta VoIP Login'. It contains a 'Password' label, a text input field, and an 'Enter' button.

Hygienic Phone Home Page

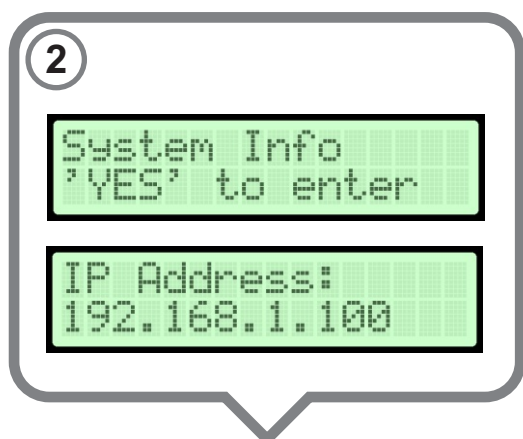
4.0 Quick Start Guide



In the web browser's address bar, enter the IP address of the phone you wish to program.

The factory default address is 192.168.1.100

Note: No password is programmed when dispatched from factory setup.



If you do not know the IP address of the phone, use the MENU and YES button on the front panel to navigate to "System Info" -> "IP Address".

Press 'MENU' once followed by 'YES' and you will see the IP Address. Press 'NO' twice to exit the menu.



Once you have successfully logged on to the phone's home page, program numbers for the call buttons.

Select 'Phone Numbers' and see the next page

4.0 Quick Start Guide

4.3 Button Phone Numbers

Here is where phone numbers are allocated to the Speed Dial numbers on the front panel. The 'Index' column is the allocated speed dial. **Please note, 10 is speed dial 0.**

001 = Speed dial 1

002 = Speed dial 2

003 = Speed dial 3

004 = Speed dial 4 (etc.)

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Call (03) 9387 7388
www.dallasdelta.com

Button Input	Name	Phone Number
001 Button 1 →	Button1	803 654321
002 Button 2 →	Button2	601
003 Button 3 →	Button3	
004 Button 4 →		
005		
006		
007		
008		
009		
010		
011		
012		
013		
014		
015		
016		

OK

Cancel

Hygienic Phone Speed Dial Assignment

DallasDelta

VoIP TELEPHONE Corporation Pty. Ltd.

Call (03) 9387 7388
www.dallasdelta.com

Name	Phone Number
Reception	900
Dispatch	901
Loading Dock	905

Enter a description in the 'Name' field of where the call will be directed.

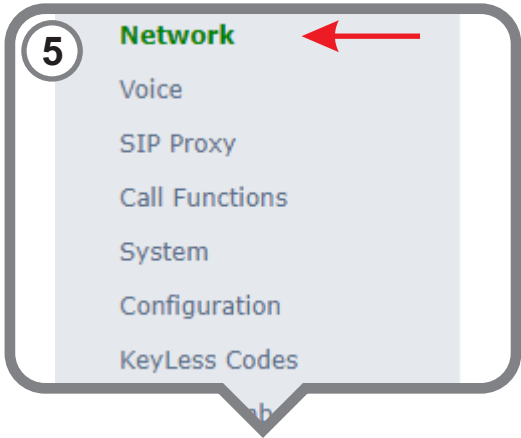
Enter the phone number you wish the button to call.

Press 'OK' to store your new entries.

4.4 Settings

4.4.1 Network Settings

Note changing the network settings at this stage will required you to re-log back into the phone if the IP is changed. If so, follow the steps outlined in 4.2 *Logging on to the Home Page*.



This page will show the current status of the registration, MAC address, and version number of the F/W.

If you phone requires a static IP address, change connection type to Static IP and fill out all the fields.

Otherwise leave the phone in DHCP mode to obtain an IP address automatically. Set “Auto DNS” when using DHCP to use the settings from the DHCP server.

Care should be taken when setting the VLAN data. Incorrect settings may require a factory default reset.

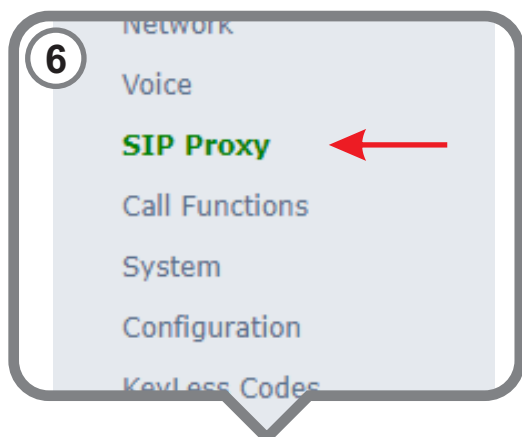
Basic Information
Phone Model: DDC_VoIP-hf
MAC Address: 38:b1:9e:df:ff:ff
Version No: 0319.11
SIP Registered : Yes > Server 1
SiteTemperature :~ 19 (DegC)

Network Settings
Connection Type:
IP Address:
Subnet Mask:
Default Gateway:

DNS Server IP:
Primary DNS:
Secondary DNS:

Layer 3 QoS: (DSCP Diff-Serv or Precedence value)
Layer 2 QoS: 802.1Q VLAN Tag
 802.1p Priority Value

4.0 Quick Start Guide



4.4.2 SIP Proxy Settings

Set up your SIP Proxy details here.

Select 'SIP Proxy' on the side menu to open up the SIP Proxy settings page.

To setup a basic SIP Registration you will need details setting from the site SIP PBX. The PBX needs to be configured for a 3rd party SIP profile. Then fill in the minimum required fields to complete the registration as follows:

SIP Server 1:

SIP Domain 1:

SIP User ID_Name:

SIP Authentication ID:

SIP Authentication PIN:
(This password is sometimes refer to as Digest KEY).

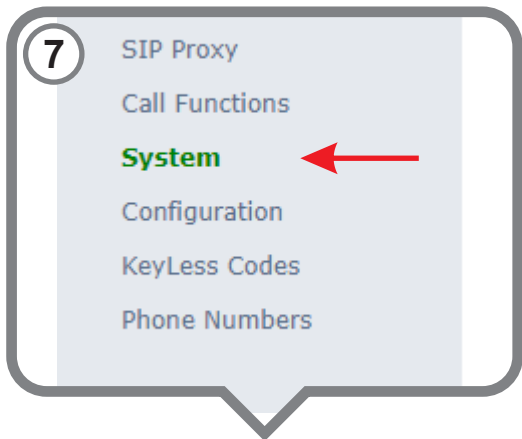
SIP Server Port:
(generally this is 5060)

All other settings are not critical at this point in time and can be set up at a later stage.

SIP Proxy Settings	
SIP Server 1:	192.168.1.90 (IP or URI)
SIP Domain 1:	192.168.1.90
SIP Server 2:	
SIP Domain 2:	
SIP Server Port:	5060 (Default 5060)
SIP Registration:	☒
Use the SIP Server As Outbound Proxy:	☐
Get SIP details from the DNS SRV:	☐
Support PRACK:	☐ (100rel, RFC3262 protocol)
SIP User ID_Name:	7123
SIP Authentication ID:	7123
SIP Authentication PIN:	 (Digest Key)
Displayed Name:	Front Door (Optional, e.g., Gate #1)
Local SIP Port:	5060 (Default 5060)
Local RTP Port (min):	6000 (RTP Starting port, default 6000)
Register Expiration:	120 (In seconds, default 120s)
Keep Alive Interval:	20 (In seconds, default 20s)

Hygienic Phone SIP Proxy Settings Page

Press 'OK' to save your settings.



4.4.3 System Settings

To access the System Settings page, you do not need to return to the Home page.

Just simply select the 'System' link on the side menu.

To setup the System for basic operation, fill in the minimum required fields to complete the registration as follows:

Relay Code 1:
This is the code used to trigger the onboard relay to open the door, gate, etc.

Note: The relay code is Factory default set to '123' and the relay contact is Normally Open (N/O).

WARNING:
For the relays to operate in gate release mode, the 'Relay Inuse' option must be set to 'None'. This can be found in the 'Phone Configuration' page. This is Factory default is 'None'.

All other settings are not critical at this point in time and can be set up at a later stage.

Press 'OK' to save your settings.

System Settings

Administration Password:

(Password to enter this web)

Handset Installed:

☐

Handset Input (0-30):

0

ACA level = 25 to 30

Handset Output (0-30):

0

ACA level = 25 to 30

Microphone Input (0-30):

15

Speaker Output (0-30):

20

Ring Volume (10-30):

15

Dial tone Vol (0-30):

15

Conversation Timer (0-99):

0

(In minutes)

Relay 1 Code:

123

eg. code to open the door/gate.

Relay 2 Code:

Relay On Timer (0-30):

5

(In seconds, Typically 5s)

Syslog IP:

255.255.255.255

Syslog time Intervals:

0

(In minutes, 0=off, max-65535 min)

Enable Debug output:

☒

SNTP Server:

0.au.pool.ntp.org

(IP or URI)
eg, 0.au.pool.ntp.org

Time Zone:

(GMT+10:00)Canberra,Melbourne,Sydney

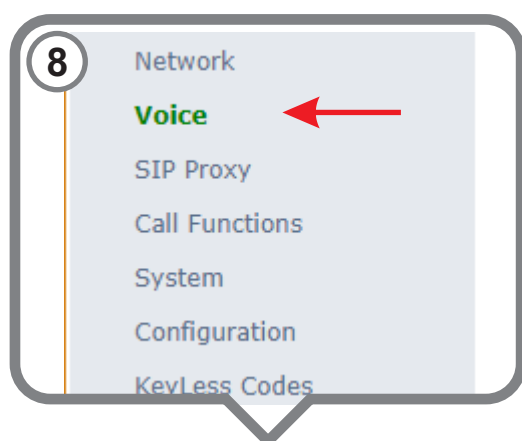
Adjust Clock for Daylight Saving:

☒

OK

Cancel

4.0 Quick Start Guide



4.4.4 Voice Codec

Most PBX phone systems today will work with the codec set to G711u (or PCMU).

This product does have other options, select preference and the order as required.

Note: The PBX can override this settings to match the other party as part of the SIP protocol. If a match can not be made the call will terminate.

Select codec priority by adjusting the selected codec in Codec 1 through Codec 5.

The “Frames per TX” option is used to adjust the number of voice frames sent per packet of voice data. A single frame is 10ms of audio. A good tradeoff between the number of packets sent versus notable audio delay is to set this to 2, in order to send 20ms of voice data per packet.

Voice Codec Settings			
Preferred Voice Codec: (In listed order)	Codec 1:	PCMU ▼	Frames per TX 1: 2
	Codec 2:	G.726-32 ▼	Frames per TX 2: 2
	Codec 3:	G722 ▼	Frames per TX 3: 2
	Codec 4:	G.729 ▼	Frames per TX 4: 2
	Codec 5:	PCMA ▼	Frames per TX 5: 2
Frames Per TX Range: Typically 1 to 4			
Send DTMF:	RFC2833 ▼		
DTMF RFC2833 Payload:	101	(Between 96 and 127, default 101)	
G.726-32 Payload Type:	111	(Between 96 and 127, default 111)	
OK Cancel			

4.0 Quick Start Guide

9



4.4.5 Reset Static IP Address (default)

The factory default address is 192.168.1.100

To set the phone to the default Static IP address, use the programming keypad on the back of the unit and enter the following key sequence:

*** # * # 7 * 1**

Note: Enter all digits in a continuous motion without pausing in between digit entries.

10



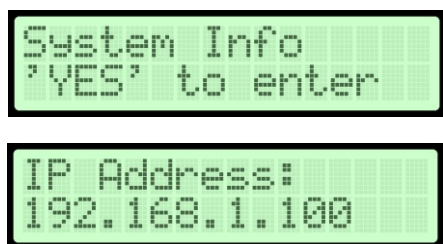
4.4.6 Reset DHCP IP Address Range

To set the phone's IP address to the DHCP range, use the programming keypad on the back of the unit and enter the following key sequence:

*** # * # 7 * 2**

Note: Enter all digits in a continuous motion without pausing in between digit entries.

11



4.4.7 Retrieving the IP Address

To check the IP address of the phone, use the MENU and YES button on the front panel to navigate to "System Info" -> "IP Address".

Press 'MENU' once followed by 'YES' and you will see the IP Address. Press 'NO' twice to exit the menu.

10



4.4.8 Reset the phone back to factory defaults

To reset the phone's parameter back to a clean start (factory reset). Enter the following key sequence:

*** # * # 7 * 9**

Note: This will reset the IP address to static.
Clear the Admin PIN. and
Erase the SIP proxy details.

5.0 Operational User Guide

The hygienic telephone operates in hands free mode.

When a speed dial button is pressed, the phone will establish a call to a preset number.

A call duration period may be set, if a limit is required.

The unit can be configured to auto-answer a call or redirect to a forwarding number the call has not been answered within the preset period.

Whilst ringing, the unit can be answered by pressing any button on the front panel.

If the called party is busy or if the number is not answered, the call can be re-direct to up to 2 alternate numbers. Refer to the 'Call Function Section- **Divert Numbers**'.

Relay Control

Two on board relays may be fitted to the hygienic phone, each may be switched on during conversation by the remote operator or either one can be configured to be activated when the call is initiated.

Remote relay control:

The hygienic phone is currently shipped fitted with one relay as standard. The second relay is installed on request.

The remote operator may activate the on board relay(s) by entering the code set for that relay. For example, if relay 1 code is set to 95, then remotely typing 95 will activate it, (refer to page 21).

A response beep will indicate that the code was correct and the relay has activated. No response tone indicates an error in the code and the process should be repeated after a minimum 3 second delay.

(NB, as relay 2 is optionally fitted, the response will be the same whether the relay is installed or not.)

DTMF remote control uses RFC2833 or INFO protocol.

Call initiated relay control:

Either one of the relays may be configured to switch on from when the call button is pressed or from when it starts ringing.

This function may be used to control an external camera, light or to initiate an alarm, etc.

The relay will stay on for the duration of the call.

(refer to **Relay Inuse** in the Configuration Section).

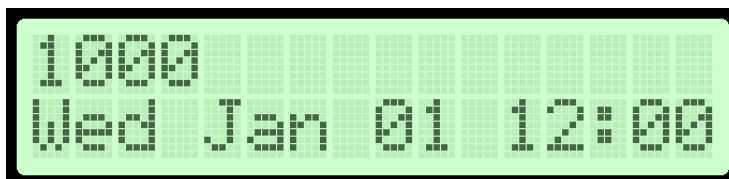
WARNING

This telephone can not be used for emergency purposes during power failure unless fitted with a backup 12 volt battery AND network connection guaranteed.

To be installed and maintained by authorised service personnel only.

5.0 Operational User Guide

LCD Display



The LCD is used to inform the user of the current status and operation of the hygienic telephone. On the idle screen, the current phone number is displayed (in the example above, it is 1000).

During a call it will show call duration, and the number dialled.

Menu Functions

Using the 'MENU', 'YES' and 'NO' buttons below the LCD you can navigate through a simple to use menu system. It provides basic information about the current status of the phone, including the IP address and other network settings, the model and current firmware version.

It also provides the ability to change between static or DHCP network mode, to aid with initial setup.

6.0 Programming Instructions

PROGRAMMING

Login page

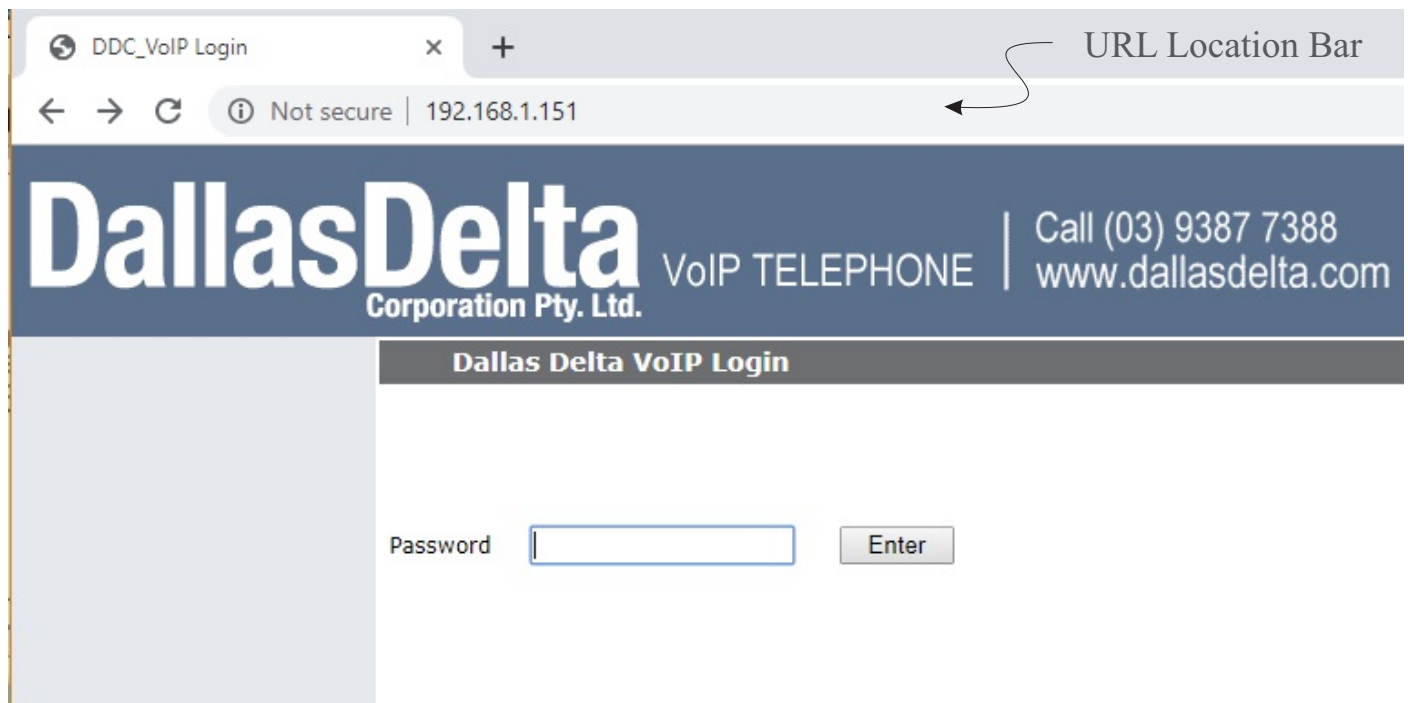
The hygienic phone is programmed by logging into the phones web page.

Use a modern web browser (Chrome or Firefox for example), enter the phones IP address into the “URL Location Bar” at the top of the screen.

If you do not know the IP address of the phone, then refer to page 14 on ‘**Retrieving the IP Address**’.

At the login page, type in the password and click the ‘Enter’ button.

By default units are delivered with no password.



6.0 Programming Instructions

SETTING MENU

On entering this section, the current status of the telephone will be displayed.

The SIP Register is updated every 8 seconds.

Submenu's are shown on the left of the WEB page, the first section is for 'Network settings' and will enable you to change IP address, subnet mask and gateway and option to set DNS server setting. The other sections are describe in the following pages of this manual.

The screenshot shows the DallasDelta VoIP Telephone web interface. The header includes the DallasDelta logo, 'VoIP TELEPHONE', and contact information: 'Call (03) 9387 7388' and 'www.dallasdelta.com'. On the left is a sidebar menu with options: Network (highlighted), Voice, SIP Proxy, Call Functions, System, Configuration, KeyLess Codes, Phone Numbers, and Contact Us. The main content area is divided into two sections. The 'Phone Status' section displays: Phone Model: DDC_VoIP-hf, MAC Address: 38:b1:9e:d0:00:00, Version No: 0319.11, SIP Registered : Yes > Server 1, and SiteTemperature : ~ 21 (DegC). The 'Network Settings' section contains fields for Connection Type (DHCP), IP Address (192.168.1.120), Subnet Mask (255.255.255.0), Default Gateway (192.168.0.1), DNS Server IP (Auto DNS), Primary DNS (192.168.0.1), and Secondary DNS. Below these are three input fields for Layer 3 QoS, Layer 2 QoS, and Layer 2 QoS, each with a description: (DSCP Diff-Serv or Precedence value), 802.1Q VLAN Tag, and 802.1p Priority Value. At the bottom are 'OK' and 'Cancel' buttons.

Network Settings

The **Connection Type** enables the unit to be connected via a *STATIC IP* or *DHCP* address mode.

If the connection is DHCP then the IP address, Mask and Gateway are automatically assigned by the server.

When set to a Static IP, a suitable IP address, Subnet Mask and Gateway address will be required to suit the local area network (LAN).

If you change the IP address you are then required to restart by logging in using the new IP address.

In most cases the DNS Server IP can remain set to **Auto DNS**, in this mode the DHCP sever will assign the DNS automatically. If however, it is required, enable **Manual DNS** and set the Primary and Secondary DNS address.

If the network switch provides QoS traffic control, then set the **Layer 2 & 3 QoS** values to match.

Set the VLAN tag option for networks with VLAN option enabled. Take care here as once this is set then the PC/Laptop you are using may not have the VLAN option enabled. You will not be able to communicate with the phone.

6.0 Programming Instructions

Voice

In this section, select the order and preferred codecs, plus the amount of frames per packet to sent for each codec type.

Also, select the options that effect the different codecs used and DTMF sending method and payloads.

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VoIP TELEPHONE

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Network

Voice

SIP Proxy

Call Functions

System

Configuration

KeyLess Codes

Phone Numbers

Contact Us

Voice Codec Settings

Preferred Voice Codec: (In listed order)

Codec 1:	PCMU	Frames per TX 1:	2
Codec 2:	G.726-32	Frames per TX 2:	2
Codec 3:	G.729	Frames per TX 3:	2
Codec 4:	G722	Frames per TX 4:	2
Codec 5:	PCMA	Frames per TX 5:	2

Frames Per TX Range: Typically 1 to 4

Send DTMF: RFC2833

DTMF RFC2833 Payload: 101 (Between 96 and 127, default 101)

G.726-32 Payload Type: 111 (Between 96 and 127, default 111)

OK

Cancel

When selecting the codecs consider the data rate that will be required by your network. As a guild the list here show the data bit rate required for each codec, if the network is via a low speed ADSL then the PCMu/a may not be a suitable choice.

Table 1. Codec comparison

<u>Codec</u>	<u>bit rate (kbit/s)</u>
G.711(PCMu /a)	64
G726-32	32
G.729	8
G.722	32

The **Frames per TX** selects how many blocks of voice data (10mS segments) are transmitted with each packet sent over the network, typically 2 - 4 is recommended.

The **Send DTMF** selects how the unit will send *DTMF* digits over the network. There are 2 protocols available, RFC2833 and INFO.

The remaining options set the payload required for each protocol and it is recommend that the default values be used.

6.0 Programming Instructions

SIP Proxy

The parameters on this page will identify the SIP server and SIP authentication details. Your ISP or network administrator will provide the necessary information to be used here.

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Network

Voice

SIP Proxy

Call Functions

System

Configuration

KeyLess Codes

Phone Numbers

Contact Us

SIP Proxy Settings

SIP Server 1:

192.168.1.90

(IP or URI)

SIP Domain 1:

192.168.1.90

SIP Server 2:

SIP Domain 2:

SIP Server Port:

5060

(Default 5060)

SIP Registration:

☒

Use the SIP Server As Outbound Proxy:

☐

Get SIP details from the DNS SRV:

☐

Support PRACK:

☐

(100rel, RFC3262 protocol)

SIP User ID_Name:

7123

SIP Authentication ID:

7123

SIP Authentication PIN:

••••

(Digest Key)

Displayed Name:

Gate 21

(Optional, e.g., Gate #1)

Local SIP Port:

5060

(Default 5060)

Local RTP Port (min):

6000

(RTP Starting port, default 6000)

Register Expiration:

1200

(In seconds, default 120s)

Keep Alive Interval:

20

(In seconds, default 20s)

Registration Attempts:

0

(before phone restarts, default 0)

Proxy Require:

NAT Traversal:

Disabled

NAT IP:

STUN Server:

(IP or URI)

STUN Server Port:

(Default 3478)

OK

Cancel

SIP Server1 will be the main running location and the **SIP Server2** can be the fail over location if available. The User detail will be the same in both servers

When **SIP registration** is required with the server then enable this option. If the phone is configured to dial a direct IP location and no SIP server is used then leave this option disabled.

If the server is via an Outbound Proxy, then use the **SIP Server** field for this proxy and enable the option **Use SIP Server as the Outbound Proxy**.

The user ID's, and PIN are supplied by the ISP/Network administrator, this information is case sensitive and should be typed in correctly. The **Displayed Name** is relayed to the connecting IP telephone as a caller ID and is optional. (Not all PBX relay this field).

SIP and **RTP** port numbers are sent to the router/firewall and are use to provide paths for voice and control data to and from the phone.

The **Register Expiration** and the **Keep Alive** timers keep the SIP server mapping and NAT ports open. The **Registration Attempts** sets how many failed attempts before rebooting the phone, this maybe preferred to re-obtain a new IP from the DHCP server and start from fresh.

If a *Provisional Response Acknowledge PRACK* is required within your network, then enable the **Support PRACK(100rel,RFC3262)** option.

When network server is behind a NAT, enable the **NAT Traversal** and set either the **NAT IP** or **STUN server** address and port.

6.0 Programming Instructions

Calling Functions

This section controls the way the phone makes and receives calls.

	Calling Functions	
Network	Forward-to Number:	(For Incoming Calls)
Voice	Forward Unconditionally:	☐
SIP Proxy	Forward When Busy:	☐
Call Functions	Forward When No Answer:	☐
System	Incoming, No Answer Period:	10 (In seconds, default 60s)
Configuration	Auto Answer:	☒
KeyLess Codes	Dial Prefix:	(Prefix digits added to Button numbers)
Phone Numbers	Hot Line Number:	(Auto-dialed number on handset pickup)
Contact Us	Divert Number 1:	(For Outgoing Calls, on Busy or No Answer)
	Divert Number 2:	
	Calling, No Answer Period:	0 (In seconds, default 10s)
	Divert for buttons only:	☐ (Only divert calls when a button is pressed)
	Dialling Timeout:	5 (In seconds, default 5s)
	Use "#" To Call:	☒

Forward to Number will set the telephone number that will be called when calls are redirected by one of three methods:

- 1) if the **Forward unconditionally** is set to yes (any calls made to the phone will be immediately redirected).
- 2) **When on busy** is set, then any attempt to call the unit when the phone is in use will be redirected.
- 3) Any un-answered call will also be redirected to this number when the **Forward When No Answer** switch is set and the **Incoming, No Answer Timeout** period has elapsed. (The Auto-answer option needs to be disable, for this to function).

The **Incoming, No Answer Timeout** period is also used the in conjunction with the **Auto Answer** switch. Set this option to 0 if the phone is not required to auto-answer.

The digits in the **Dial Prefix** are added to the any number dialled via the keypad or the preprogrammed button numbers.

When the hygienic phone is making a call, it is possible to divert the call to 2 alternate numbers if the other party is busy or doesn't answer. To enable this function, enter the **Divert Numbers** and set the **Outgoing, No Answer Period** a value in second.

The **Divert for buttons only** is used if your unit has both buttons and a keypad installed.

The **Dialling Timeout** period sets how long to wait after the last digit pressed before dialling the number. The other option is to enable the **Use # to Call** to initiate dialling (only when dialling via a keypad).

6.0 Programming Instructions

System Settings

Set the hygienic phone audio levels, relay options, Syslog details and time of day functions are adjusted in this section. Audio levels for the phone should be re-check on site.

The on board relay(s) may be used for gate or door/area access control, back ground lighting or camera activation. They can be remotely activated or enabled when the hygienic phone is in use.

System Settings	
Administration Password:	(Password to enter this web)
Handset Installed:	☒
Handset Input (0-30):	20 ACA level = 25 to 30
Handset Output (0-30):	20 ACA level = 25 to 30
Microphone Input (0-30):	15
Speaker Output (0-30):	20
Ring Volume (10-30):	15
Dial tone Vol (0-30):	15
Conversation Timer (0-99):	0 (In minutes)
Relay 1 Code:	123
Relay 2 Code:	321
Relay On Timer (0-30):	5 (In seconds, Typically 5s)
Syslog IP:	255.255.255.255
Syslog time Intervals:	0 (In minutes, 0=off, max-65535 min)
Enable Debug output:	☒
SNTP Server:	0.au.pool.ntp.org (IP or URI) eg, 0.au.pool.ntp.org
Time Zone:	(GMT+10:00)Canberra,Melbourne,Sydney
Adjust Clock for Daylight Saving:	☒

The **Administration Password** is used to gain access to the web page. For added security set and document this password. The phone is supplied with no password set.

The microphone and the speaker levels for the handset (if fitted) and in the hands-free circuit, is adjusted to suit site background noise, when possible keep these level low, as un-necessary noise will be transmitted back to the remote operator.

If the phone does have a handset installed, then enable the option and set the audio levels as required for the handset.

Set the **Ringer Volume**, **Dial Tone** levels to suit site conditions.

Conversation period may be limited if desired, to set this, enter a period in minutes into the **Conversation Timer** field. If no timer is required then enter a 0 to disable this function.

The onboard relay may be activated remotely when the remote operator dials the digits that matches the fields for **Relay 1 & Relay 2 code**. Note that relay 2 is optionally fitted.

Use the **Relay on timer** field for the duration required. Or set it to 0, to remain on for the duration of the call.

Debugging can be enabled by Syslog server and/or UDP. If this is not required then disable these options. refer to the next page for more details on these option

NOTE: ACA maximum levels:

Do not exceed the handset maximum levels, for conformity and comfort keep handset levels to a minimum.

6.0 Programming Instructions

System Settings continued

Debug Options:

For some installation, it may be required to record events that occur at the phone. These events may include;

- which button is pressed
- when a relay is activated remotely
- when the handset is on or off hook
- on ringing (incoming call)
- system reboot
- start of call and at auto answer
- at the end of a call
- and a status of it's current state, ie Logged-on or not at set intervals.

This information is stored at an allocated server and the protocol used is *SYSLOG on port 514*. Although generally on a Linux systems, *SYSLOG* programmes can be sought for the Windows system as well.

Each event is timed stamp with the phone ID and IP address. To enable this function set the IP of the *SYSLOG* server and the intervals.

The hygienic phone outputs general debugging information on a continuous basis on port 8225. If high traffic is a concern then disable the **Enable Debug output** option.

The 'time of day' used within the phone and for Syslog is set either from the SNTP server or the SIP server. If the LAN does not have access to a SNTP server then set this field to 0.0.0.0

The **SNTP Server** (simple network time protocol) provides 'time of day' for syslog events. SNTP is on port '123'. If the phone is connected to the internet then a site like 'time.windows.com' or '0.au.pool.ntp.org' may be used.

Select the State/ Country in the **Time Zone** option and whether to use '**DayLight saving**' time adjust is required in the next 2 fields

6.0 Programming Instructions

Configuration

This configuration section control some of the hardware and functions of the phone.

The '**Relay In Use**' will allow either relay to switch on whenever the phone is being used. It will set the relay on when making and receiving a calls and stay on for the duration of the call.

Enable '**Soft Ramp Ringer**' to have the incoming call ring volume ascend slowly over time as to reduce the chance of shock to nearby persons performing delicate or careful operations.

Phone Configuration	
Key '#' Function	Normal
Relay Inuse	None
KeyPad Speed-Dial:	Disabled Memory Dial on first or second key-press:
Dial Numbers Silently:	☒ (When a Button is pressed or Speed Dialling)
Disconnect After Gate open:	☐
SNMPv2c Trap Destination_1:	192.168.1.171 (SNMPv2c Traps server IP address)
SNMPv2c Trap Destination_2:	0.0.0.0
SNMPv2c Community Name:	DDC_VoIP
Auto Provisioning Server:	192.168.1.90/provisioning (IP or URI)
Auto Provisioning Port:	80 (Default 80)
Auto Provisioning Interval:	100 (In minutes, 60 to 65535 minutes)
Temperature calibration:	-4 (Offset +/-10)

When a call is made via pressing a Speed-Dial key, the tones of each digit may be muted, which is useful for long phone numbers. If this suits, then enable the '**Dial Number Silently**' option.

In addition to the other debugging options, this product also can send SNMP traps to a server. It uses v2c protocol, If used then this protocol requires your community name as well as one or two servers IP addresses. It has an enterprises code of 1.3.6.1.4.1.45255.1.1.1.0 to .10.

A typical Get command would be '**snmpget -v2c -c DDC_VoIP 192.168.1.100 1.3.6.1.4.1.45255.1.1.1.0**'.

The **Auto Provisioning Server** maybe set up if the company has many phones on the network, allowing the admin officer to make changes to the functions of the phone as an alternative method to accessing via the web page. Generally, for few phones the web access is an easier method.

The Temperature output seen on the Status section can be offset by adding to the '**Temperature Calibration**' value. Note that this value is only an approximation of +/-5 DegC, and only to be used as an indication of the site temperature.

6.0 Programming Instructions

Keyless Codes

Network

Voice

SIP Proxy

Call Functions

System

Configuration

KeyLess Codes

Phone Numbers

KeyLess Codes

Keyless Entry Codes:

123

456

789

"To access Keyless entry at the gate, user will type * plus one of the above codes"

OK

Cancel

Keyless codes are typically used to allow entry from one area to another. By entering the keyless entry code in the front panel a relay will be activated. This will only activate relay1 of the phone. The period that the relay stays on for is the same as that set in the relay option section. (refer to page 22). The user is required to enter * and the code to open the gate. For areas that require an Emergency Services access, then it maybe advisable to use a code that is 8 to 10 digits long, as it may not be changed that easily once the code is supplied to the Emergency services personal.

6.0 Programming Instructions

Phone Numbers

The hygienic phone has 10 speed dial buttons on the front panel.

Each button may be set to dial a different phone number. The Name field given to each input is for internal reference only and is not required.

The phone number may be an IP address/port number (i.e. 192.168.1.123:5060) if the unit is not registered to a SIP server.

Each number maybe set to a URI location like '912345678@mysipproxy.org'

	Button Input	Name	Phone Number
Network	001	Button1	903 654321
Voice	002	Button2	901
SIP Proxy	003	Button3	192.168.1.56 93877388
Call Functions	004		
System	005		
Configuration	006		
KeyLess Codes	007		
Phone Numbers	008		
	009		
	010		
Contact Us	011		
	012		
	013		
	014		
	015		
	016		

Special commands within the phone number,

The hygienic phone can be configured to connect to a PSTN/PBX via a ATA. In this scenario the unit calls the ATA and then, on answer, the phone then dials the PBX number.

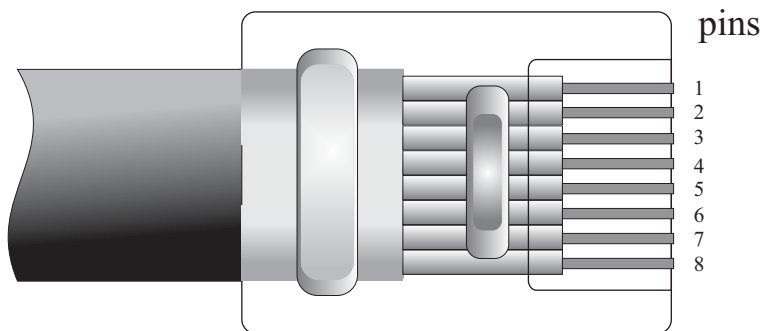
To do this, enter the number or the IP address of the ATA, followed by a pipe character '|', and then PSTN telephone number to call. (the PBX number may include a pause by adding a comma ','). Example, if the ATA number is 1234 and the PBX number is 9,91234567, then you would enter 1234|9,91234567.

Note, if all calls are made via the ATA, then the number can be loaded in the **Dial Prefix** location, In this case the Dial Prefix number would be '**1234|**'.

It may be required to dial this number without each DTMF tone played out over the speaker, especially so, for long phone numbers like an IP address, to enable silence dialling, enable the option **Dial numbers silently** (refer to page 24).

7.0 Specifications

power:	input voltage	9 Volts (<i>minimum</i>) - 50 Volts (<i>maximum</i>)
PoE supply:	Class requirements	Class 0. (0.44 Watts to 12 Watts)
current consumption:	-idle mode -on call	125mA @ 12Vdc (1.5 Watts) <250mA @ 12Vdc (3 Watts) <1300mA @ 12Vdc (15 Watts) <i>max volume into a 8ohms speaker</i>
relay contacts:	switching maximum	1A @ 60Vdc / 40Vac SELV or TNV (non inductive load) voltage free outputs
temperature:	operating range	0°C to +50°C
SPL:	ringer output level	>80dBa @ 1 metre
communication:	Ethernet Connection protocol CODECs	100 BASE-T SIP G711 (uLaw, aLaw), G726-32, G722, G729
physical:	panel dimensions (mm) wall cut-out (mm) weight (kg)	280 x 245 230 x 225 x 40 approx. 1.4



RJ45 Ethernet connector

pins		
1 = TXP	}	mode A PoE
2 = TXN		
3 = RXP		
6 = RXN		
4,5 = Spare	}	mode B PoE
7,8 = Spare		

WARNING

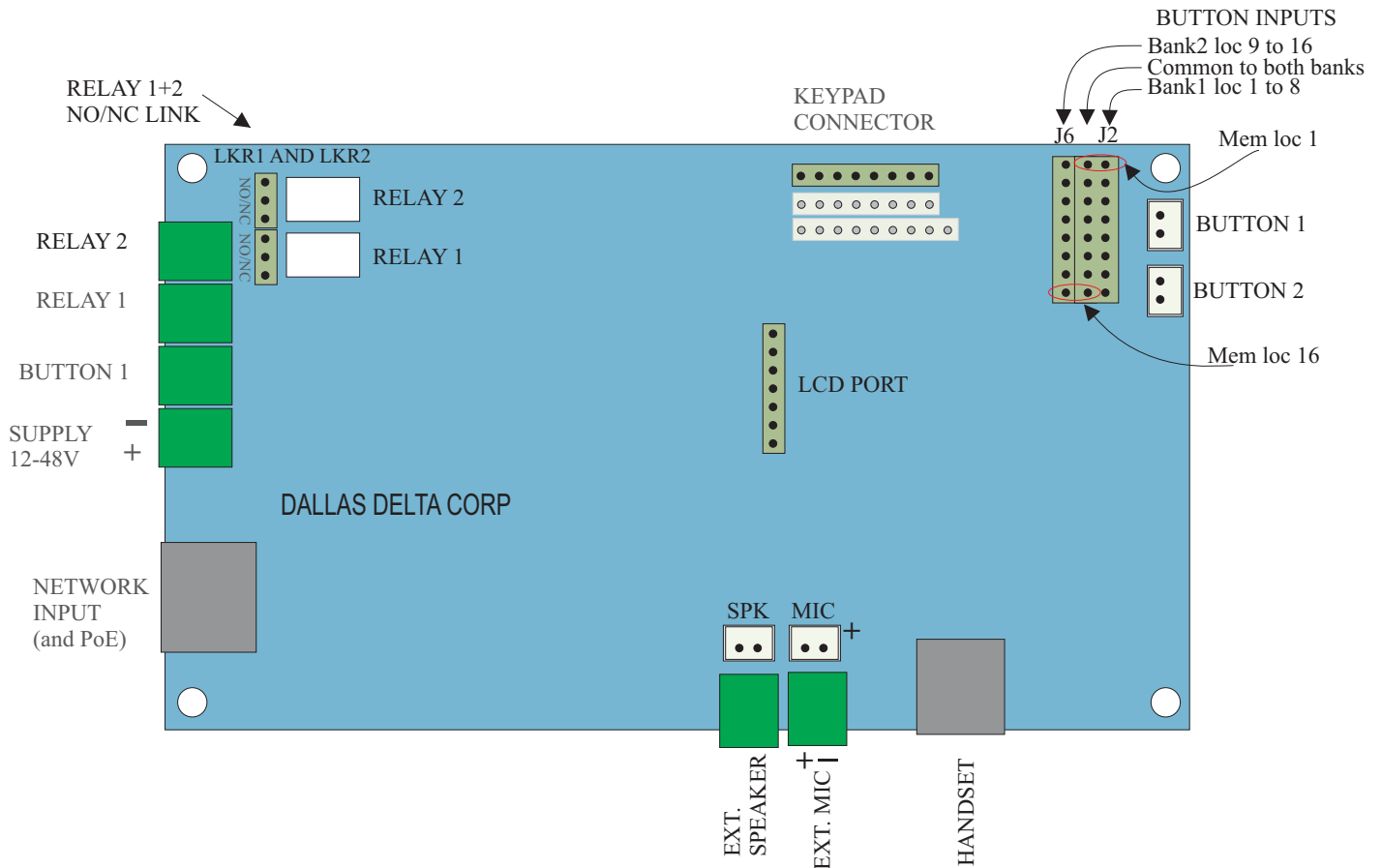
This telephone can not be used for emergency purposes during power failure unless fitted with a backup 12 volt battery AND network connection guaranteed.

To be installed and maintained by authorised service personnel only.

8.0 PCB Layout

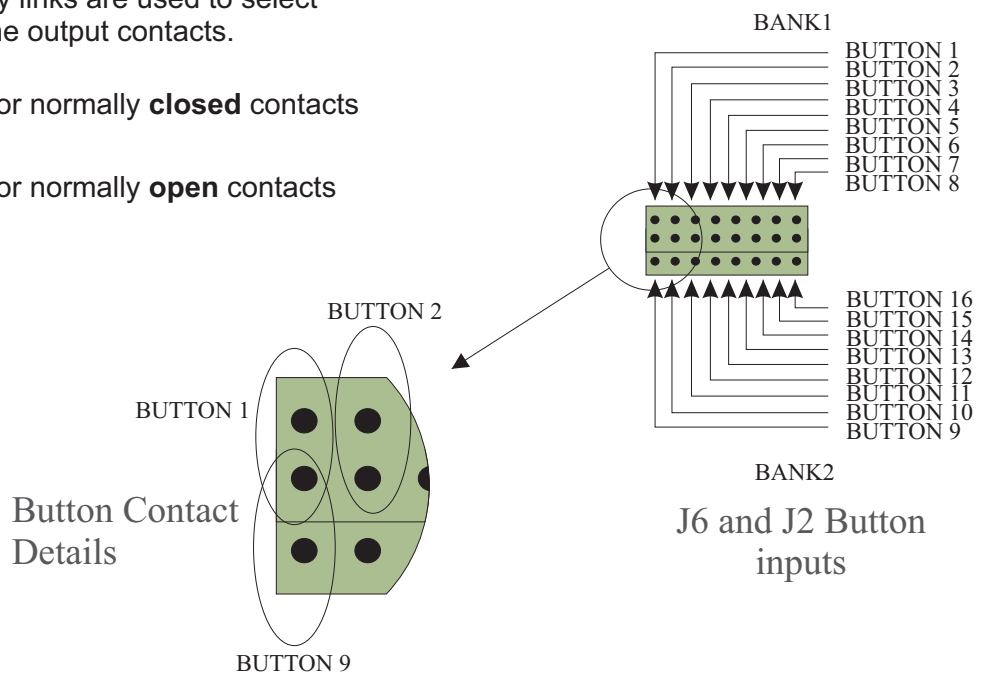
DDC_VoIP S_11 PCB Layout

The DDC_VoIP PCB connection may be different from the diagram shown, The layout is of a generic configuration, some items may not be installed and/or not required.



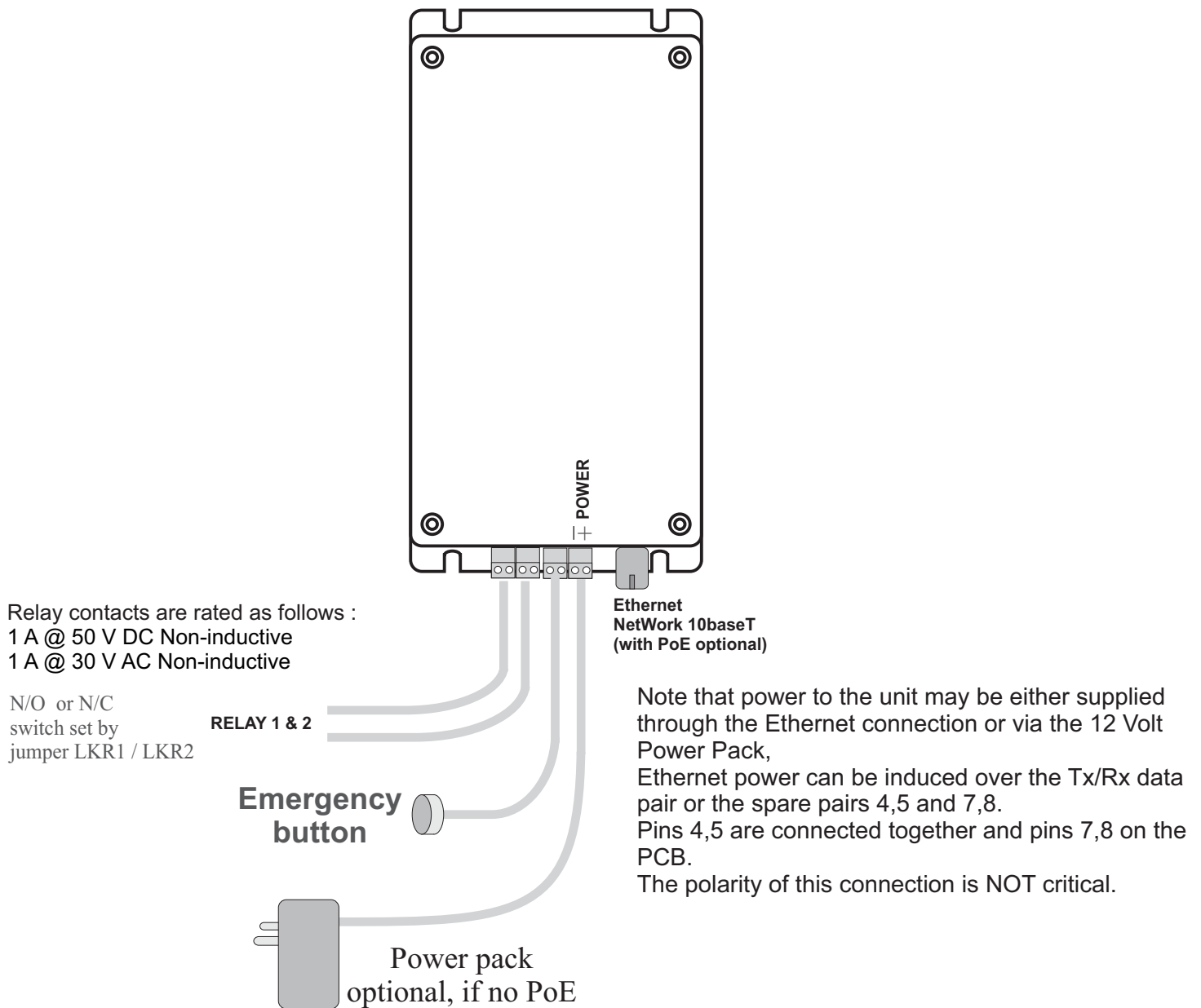
LKR1 and LKR2 relay links are used to select the normal state of the output contacts.

-  Relay linked for normally **closed** contacts
-  Relay linked for normally **open** contacts



9.0 Connection Details

ELECTRICAL WIRING



WARNING

This telephone can not be used for emergency purposes during power failure unless fitted with a backup 12 volt battery AND network connection guaranteed.

Note: All wiring is to be routed away from high EMI radiating devices, such as transformers, fluorescent lighting etc.



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